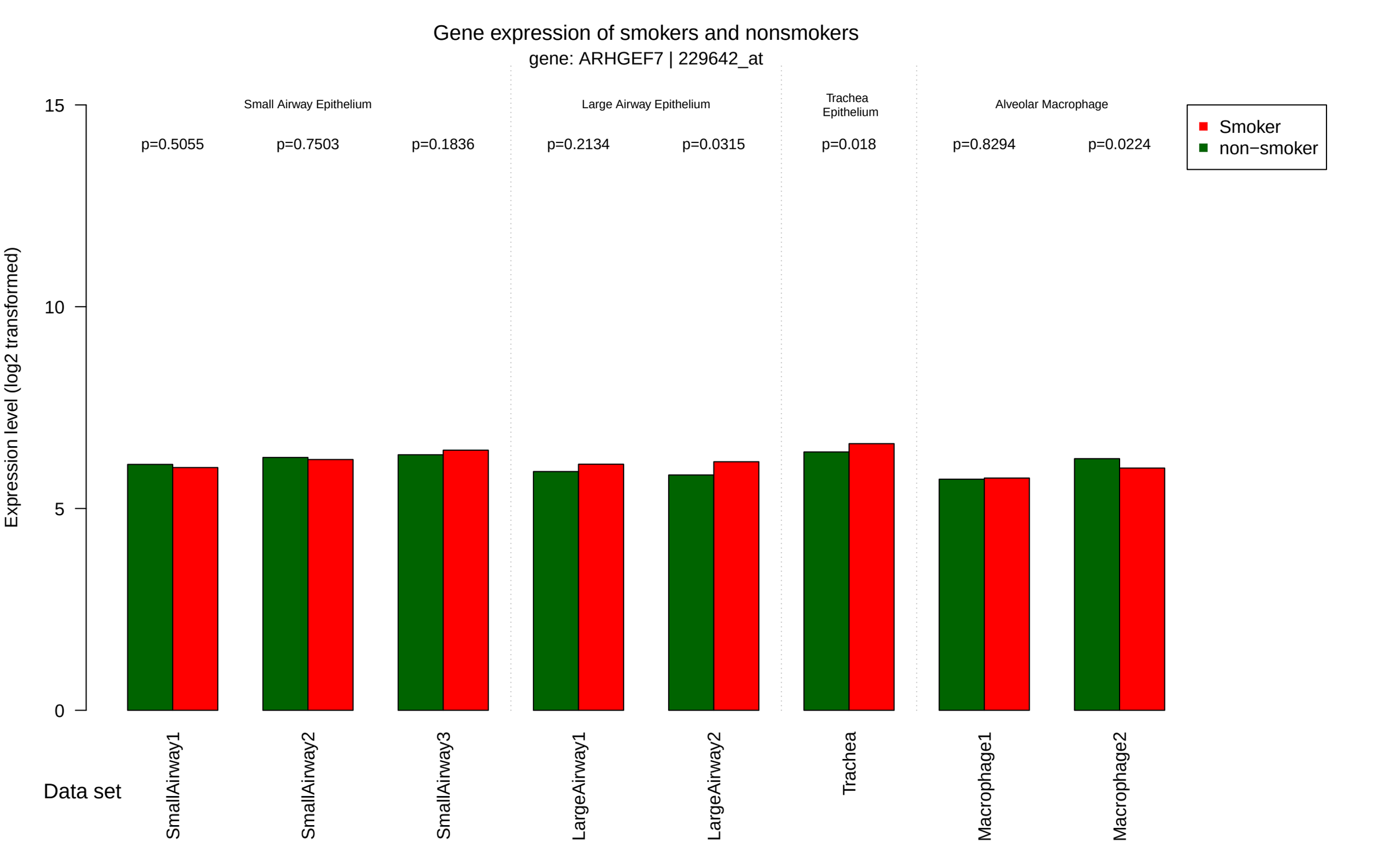
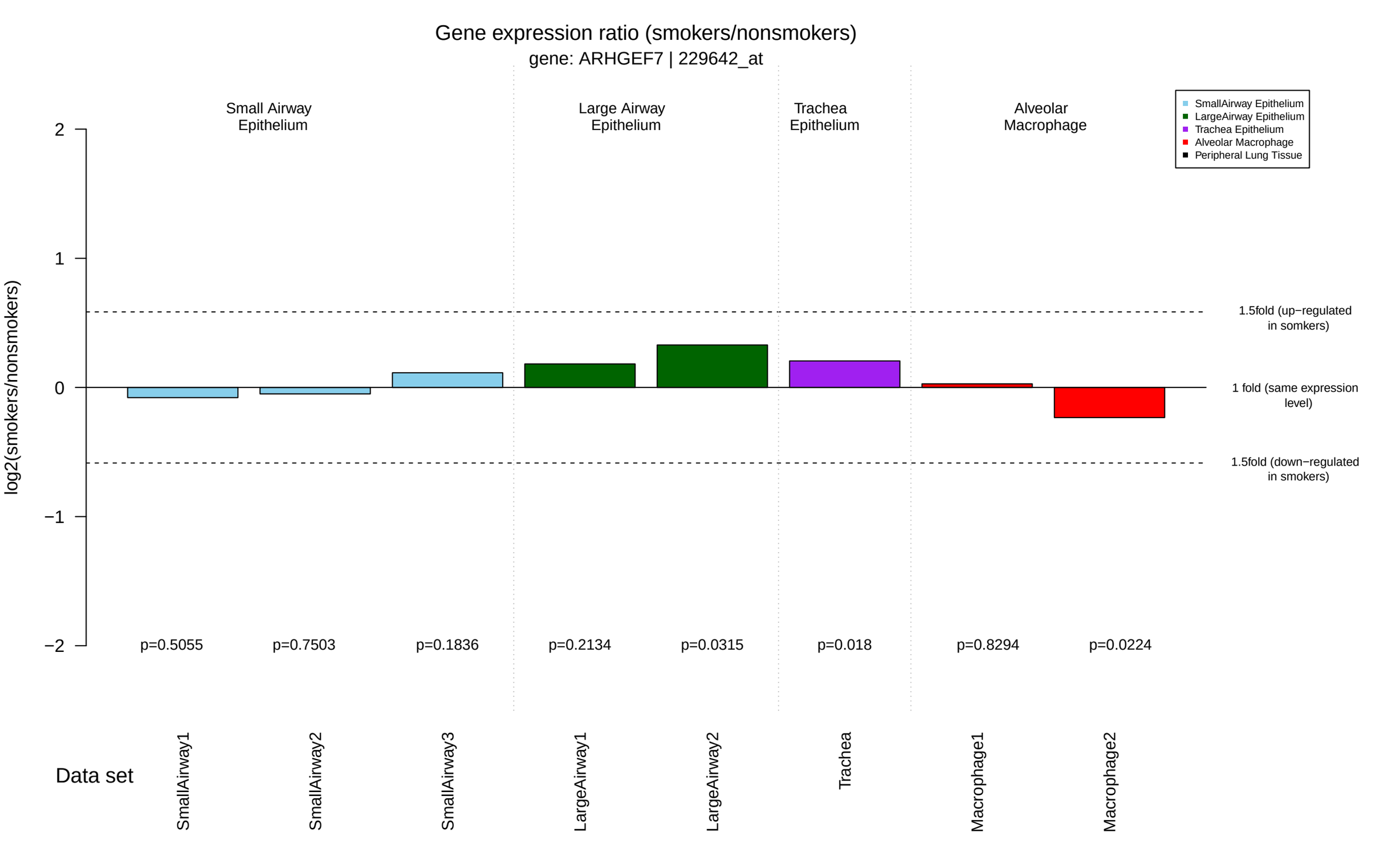
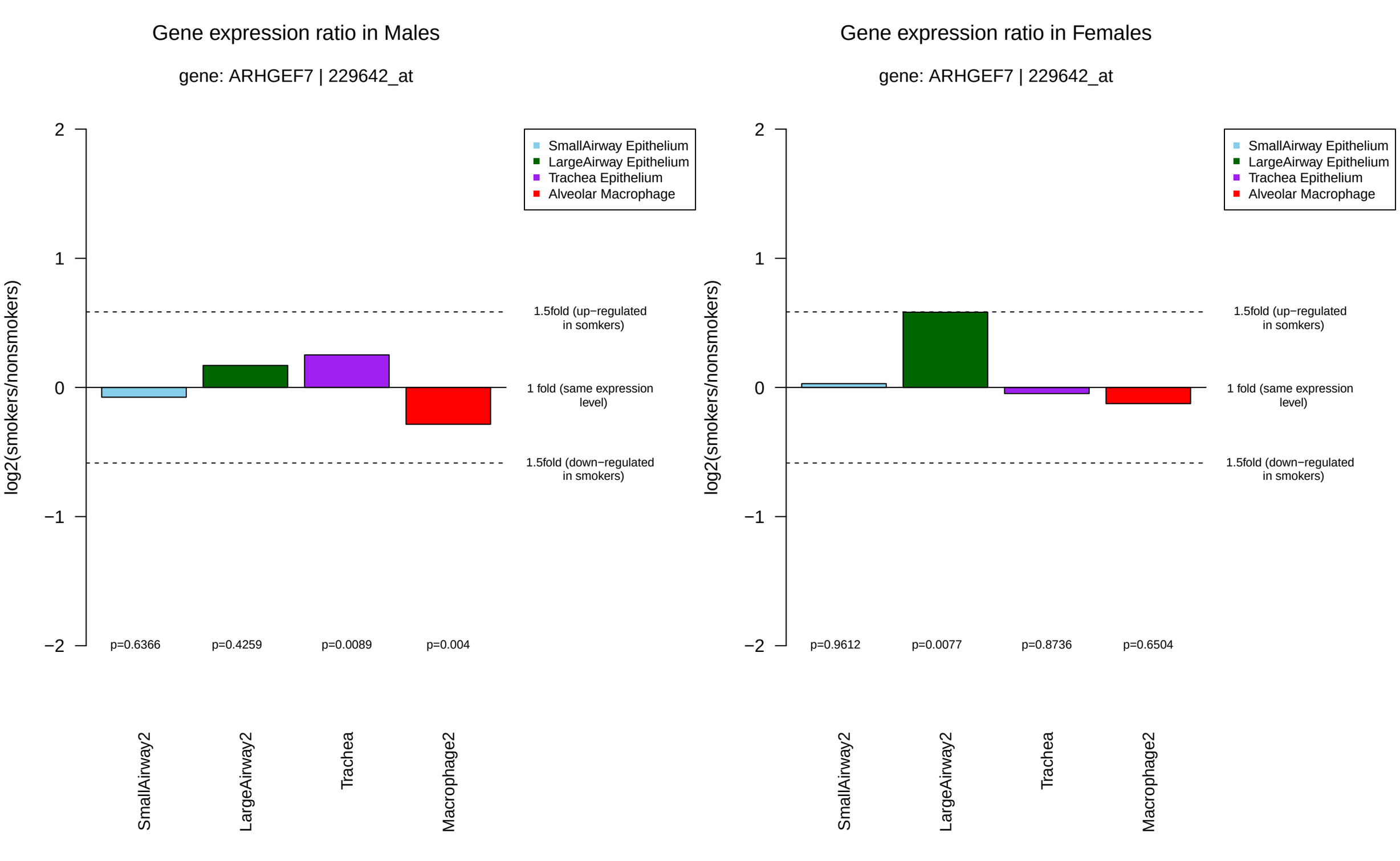
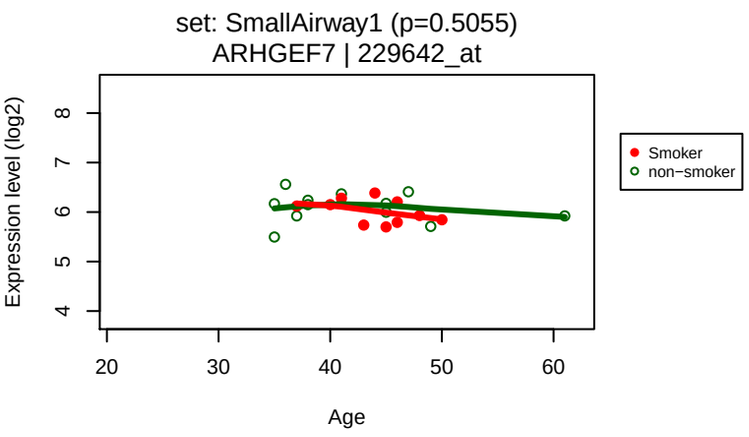
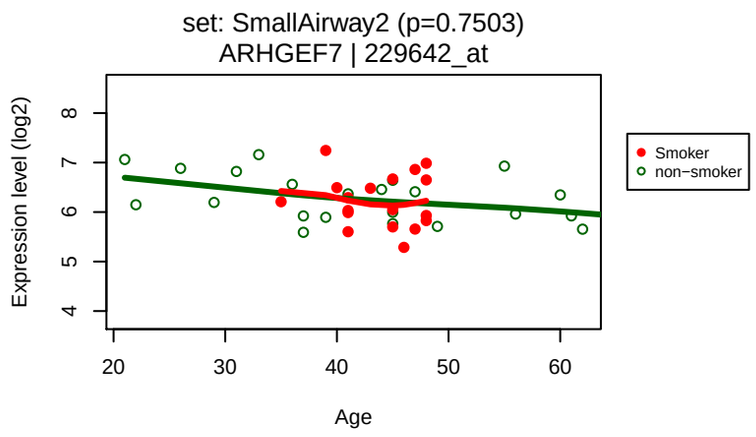
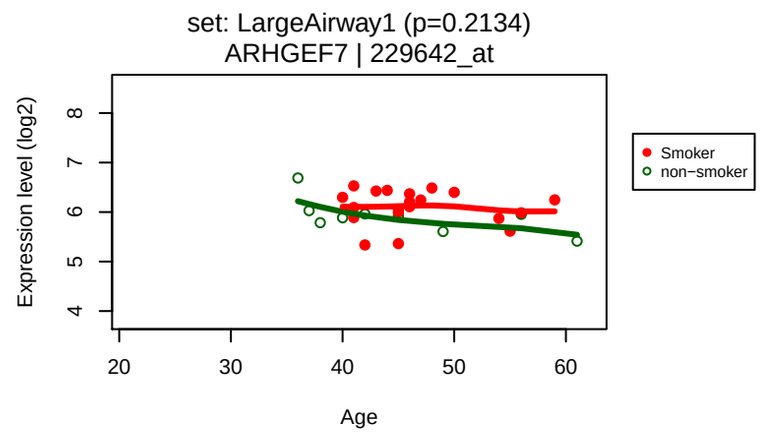

Gene expression of smokers and nonsmokers
gene: ARHGEF7 | 229642_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5055
p=0.7503
p=0.1836
p=0.2134
p=0.0315
p=0.018
p=0.8294
p=0.0224
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

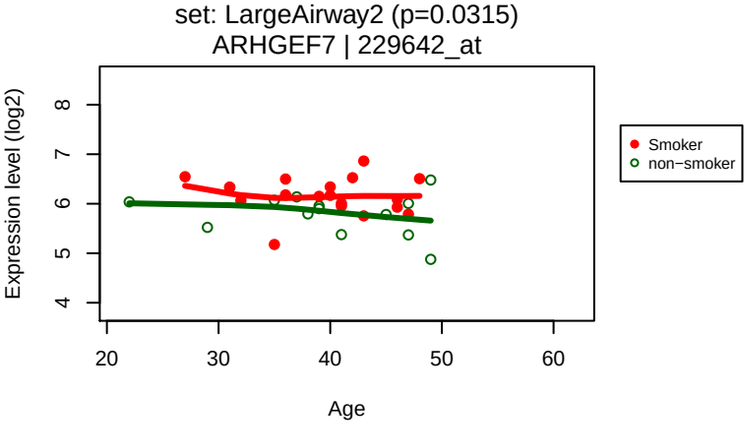

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF7 | 229642_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5055
p=0.7503
p=0.1836
p=0.2134
p=0.0315
p=0.018
p=0.8294
p=0.0224
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

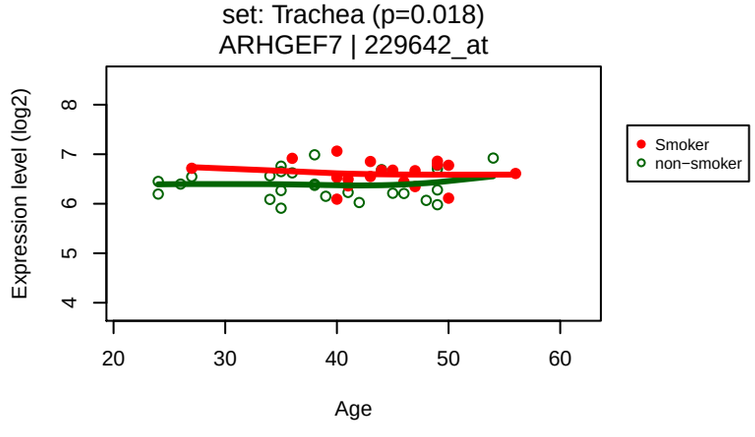

Gene expression ratio in Males
gene: ARHGEF7 | 229642_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6366
p=0.4259
p=0.0089
p=0.004
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF7 | 229642_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9612
p=0.0077
p=0.8736
p=0.6504
SmallAirway2
LargeAirway2
Trachea
Macrophage2

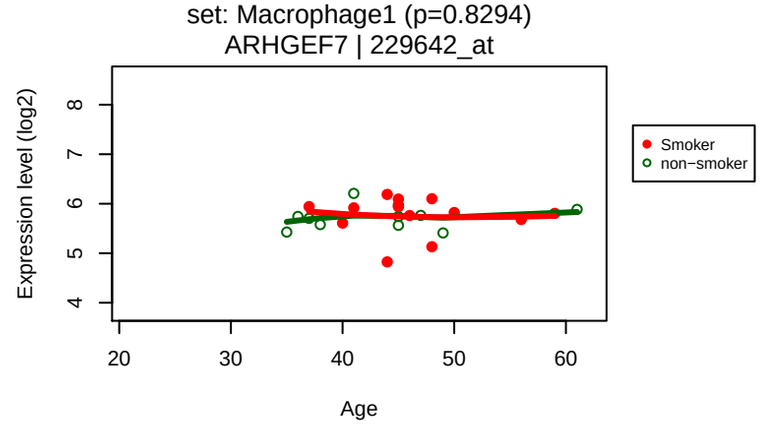

set: SmallAirway1 (p=0.5055)
ARHGEF7 | 229642_at
Expression level (log2)
Age
Smoker
non−smoker

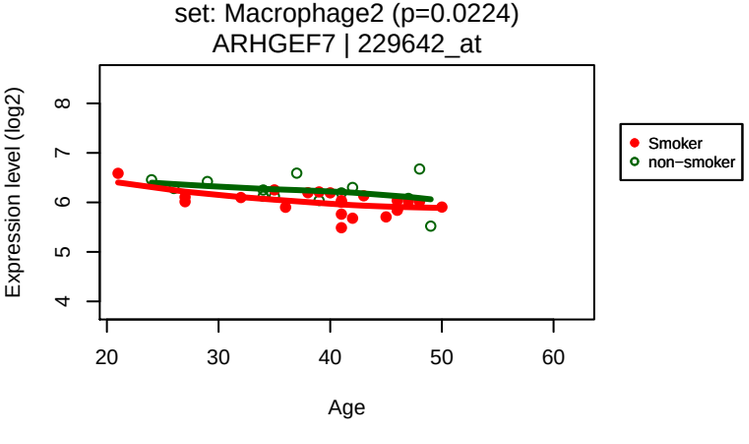

set: SmallAirway2 (p=0.7503)
ARHGEF7 | 229642_at
Expression level (log2)
Age
Smoker
non−smoker

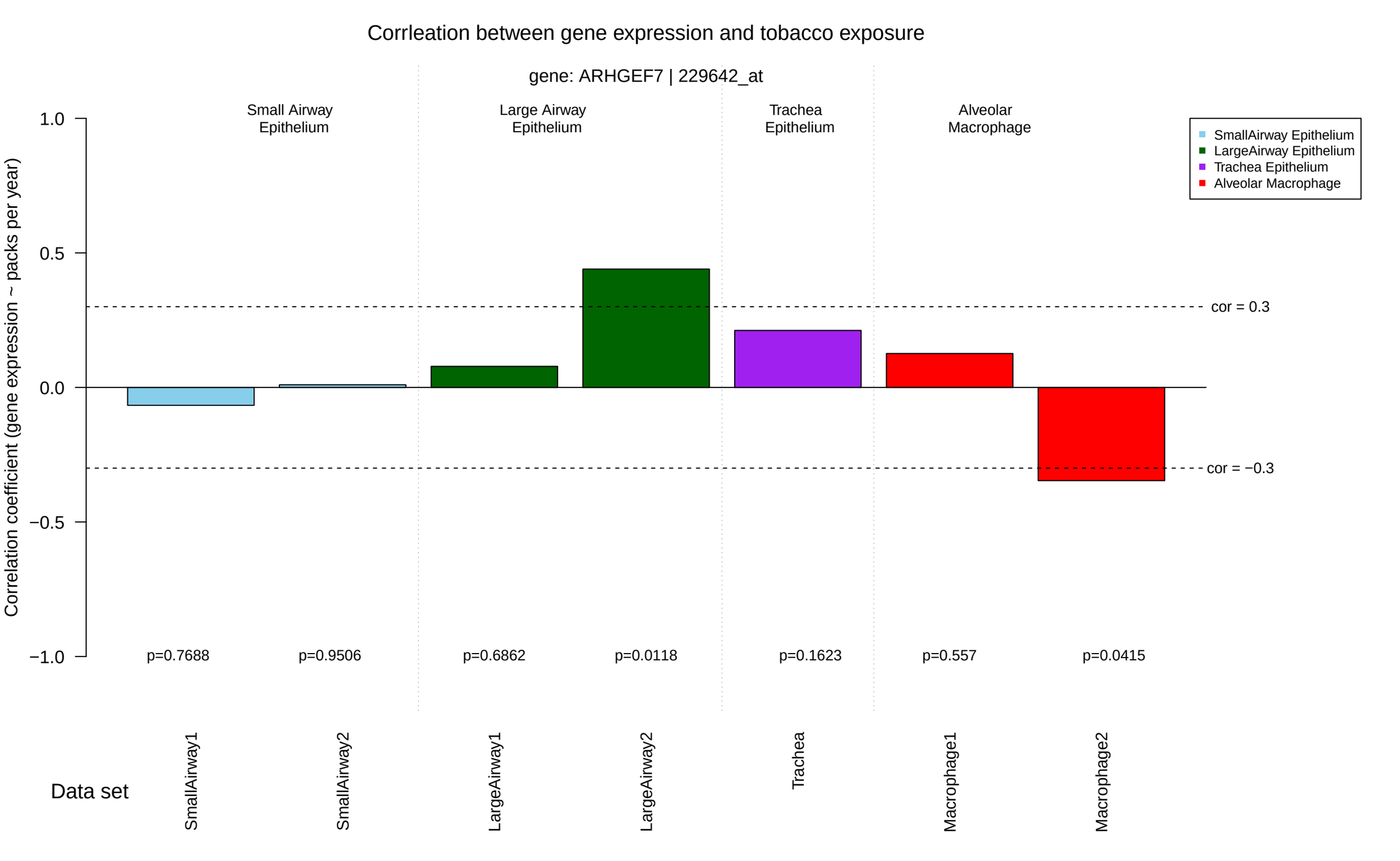

set: LargeAirway1 (p=0.2134)
ARHGEF7 | 229642_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0315)
ARHGEF7 | 229642_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.018)
ARHGEF7 | 229642_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8294)
ARHGEF7 | 229642_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0224)
ARHGEF7 | 229642_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF7 | 229642_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7688
p=0.9506
p=0.6862
p=0.0118
p=0.1623
p=0.557
p=0.0415
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2